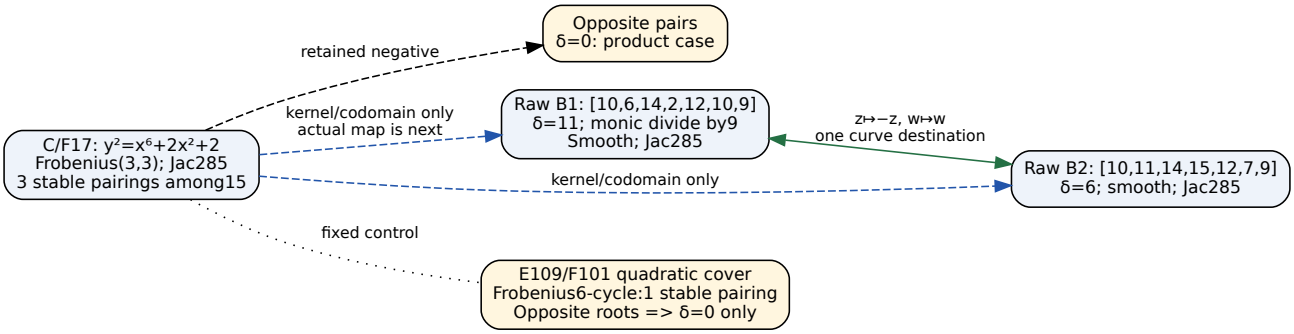
Frobenius-stable kernels: codomain assessment

2026-11-11 · Chained ECDLP Transfers · fixed F17/F101 controls

Unsigned/unverified standing. The code constructs kernel splittings and codomain models. The actual non-split divisor correspondence remains next. PCC-42713c binds development results. No benchmark or cost gain. [pcc:unverified]



Exact F17 outcome

Use predeclared E:y²=x³+2x+2,order19,G=(5,1), with C:y²=x⁶+2x²+2. In F17[T]/(T³+2T+2), choose β²=T by the first coefficient encoding. Its three conjugates and negatives give Frobenius(012)(345). All15 pairings are enumerated;3 are stable. Opposite pairs have δ0. The two cross pairings have determinants11 and6 and smooth target arrays shown above. All coefficients descend to F17. Leading9 has square root3; monic normalization divides the equation by9. Each Jacobian has285 rational classes. [pcc:unverified]

Both targets are isomorphic by z↦−z,w↦w, so they are one destination. The full4,896-matrix PGL2(F17) audit finds no curve isomorphism from source to either target. It finds2 source self-maps on the quotient line and1 for each target. This is a curve-level distinction, without an unpolarized Jacobian-isomorphism inference. Higher-degree elliptic maps remain possible. [pcc:unverified]

Exact F101 negative

For the existing E109 source y²=x³+x+46, the cubic is irreducible and a root has nonsquare norm55. Euler's criterion in its cubic field makes that root nonsquare. Its square roots give a sextic6-cycle. Only the opposite-root matching is stable; the resulting quadratics X²−αi have zero linear column and δ0. This specific quadratic cover cannot directly start a nondegenerate F101-rational Richelot hop to a genus2 codomain. [pcc:unverified]

Proof scope, deduplication and next implementation

The F101 result is a fixture-specific kernel restriction. It does not rule out shifted/other covers, other degrees, paths through elliptic products, extension fields or computational transfers generally. The earlier broad rational-factor limitation is not promoted to universal impossibility. [pcc:unverified]

Five upstream Lean certificate/accounting lemmas audit the explicit15-row matching table, its stable filters for6 and(3,3) cycles, the zero determinant column and odd-norm exponent identity. The5-declaration gate passes; crypto PCC465142 remains unsigned. Geometric kernel identification and finite-field norm theory remain written primary-source inputs. No finite field or actual Richelot map is constructed by these lemmas. [pcc:unverified]

Primary source: Kunzweiler,DCC92(2024),§§3.1–3.2,Lemma6 and Propositions7–8, DOI10.1007/s10623-024-01366-1. Its normalized determinant/brackets fix twists and signs. Galbraith corrected ch.10§10.4.2 supplies the balanced model; ch.8§8.3 supplies Picard norm/pullback. Full account: NONSPLIT_KERNELS.md. Source: src/nonsplit.rs, reusing pre-existing BigUint polynomial arithmetic.

Frozen next-map acceptance criteria

Use the first nondegenerate pairing in the complete ordering, δ11, before evaluating images. Target raw ordinate w=3Y. Combine $F17^2$ source support and $F17^3$ coefficients inside $F17^6$. Keep each target fiber as a quadratic Mumford pair, avoiding target root selection. Handle the balanced infinity terms and require coefficientwise descent to F17. [pcc:unverified]

Implement the actual correspondence and sign-corrected reversed dual, then check both dual norms[2] on every285 rational class, additivity, all19 source scalar relations and reversed-chain/source norm[4] with inverse5 modulo19. These are prospective tests. Target construction alone is not an executable ECDLP transfer. [pcc:unverified]

The two equivalent target models and degenerate source pairings remain in the record. Future chain searches must track them instead of counting each model as a new Jacobian destination. No cost, solver or prime-field descent result is inferred from a smooth codomain or preserved odd-group theorem. [pcc:unverified]

Reproduce with check.sh and the unchanged crypto75a737b8 runtime pin. Final receipt nonsplit-pinned-02.txt passes guards/formatting/strict Clippy, native25/reference40 tests and1 existing ignored benchmark. PCC-3c51f5 is unsigned. Earlier development and complete passing receipts are retained. [pcc:unverified]

No experiment/paid job launched. S3 checkpoint and live IsoLab budget gates remain unavailable; KB/atlas retrieval is blocked. Crypto’s frozen ledger fork and archived CI requirements still block PR3877. The persistent queue is crypto3904; source resume state is CHECKPOINT.md. Canonical leaderboard and transfer graphs remain unchanged: no signed or measured edge promoted.